

HDOC Day-25

Write a C program that displays the following menu and uses a switch statement to print the “Decreasing patterns” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value.

1. Decreasing Right-Angled Triangle

Algorithm

1. Start.
2. Input the number of rows n.
3. Use an outer loop from i = n down to 1.
4. For each row, use an inner loop to print * exactly i times.
5. Move to the next line.
6. Stop.

C Function

```
#include <stdio.h>
```

```
void decreasingStar(int n)
```

```
{
```

```
    int i, j;
```

```
    for (i = n; i >= 1; i--)
```

```
    {
```

```
        for (j = 1; j <= i; j++)
```

```
        {
```

```
            printf("* ");
```

```
        }
```

```
        printf("\n");
    }
}
```

2. Decreasing Natural-Number Pattern

Algorithm

1. Start.
2. Input the number of rows n.
3. Set the first row to contain numbers from 1 to n.
4. Use an outer loop from i = n down to 1.
5. Use an inner loop from j = 1 to i.
6. Print j.
7. Move to the next line.
8. Stop.

C Function

```
void decreasingNatural(int n)
{
    int i, j;

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}
```

Decreasing Repeated-Number Pattern

Algorithm

1. Start.
2. Input the number of rows n.
3. Use an outer loop with i = n down to 1.
4. Use an inner loop to print the value i, exactly i times.
5. Move to the next line.
6. Stop.

C Function

```
void decreasingRepeated(int n)
```

```
{  
    int i, j;  
  
    for (i = n; i >= 1; i--)  
    {  
        for (j = 1; j <= i; j++)  
        {  
            printf("%d ", i);  
        }  
        printf("\n");  
    }  
}
```

Reverse-Number Triangle

Algorithm

1. Start.
2. Input the number of rows n.
3. Use an outer loop from i = n down to 1.
4. Set j = i.
5. Print numbers from i down to 1.

6. Move to the next line.

7. Stop.

C Function

```
void reverseNumber(int n)
```

```
{  
    int i, j;  
  
    for (i = n; i >= 1; i--)  
    {  
        for (j = i; j >= 1; j--)  
        {  
            printf("%d ", j);  
        }  
        printf("\n");  
    }  
}
```

Complete Menu-Driven C Program

```
#include <stdio.h>
```

```
void decreasingStar(int n)
```

```
{  
    int i, j;  
  
    for (i = n; i >= 1; i--)  
    {  
        for (j = 1; j <= i; j++)  
        {  
            printf("* ");  
        }  
    }  
}
```

```
    }  
    printf("\n");  
}  
}
```

```
void decreasingNatural(int n)
```

```
{  
    int i, j;  
  
    for (i = n; i >= 1; i--)  
    {  
        for (j = 1; j <= i; j++)  
        {  
            printf("%d ", j);  
        }  
        printf("\n");  
    }  
}
```

```
void decreasingRepeated(int n)
```

```
{  
    int i, j;  
  
    for (i = n; i >= 1; i--)  
    {  
        for (j = 1; j <= i; j++)  
        {  
            printf("%d ", i);  
        }  
    }  
}
```

```
    }  
    printf("\n");  
}  
}
```

```
void reverseNumber(int n)
```

```
{  
    int i, j;  
  
    for (i = n; i >= 1; i--)  
    {  
        for (j = i; j >= 1; j--)  
        {  
            printf("%d ", j);  
        }  
        printf("\n");  
    }  
}
```

```
int main()
```

```
{  
    int choice, n;  
  
    printf("==== DECREASING PATTERNS =====\n");  
    printf("1. Decreasing Right-Angled Triangle\n");  
    printf("2. Decreasing Natural-Number Pattern\n");  
    printf("3. Decreasing Repeated-Number Pattern\n");  
    printf("4. Reverse-Number Triangle\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
printf("Enter number of rows: ");
```

```
scanf("%d", &n);
```

```
if (n <= 0)
```

```
{
```

```
    printf("Invalid number of rows.\n");
```

```
    return 0;
```

```
}
```

```
switch (choice)
```

```
{
```

```
    case 1:
```

```
        printf("\nDecreasing Right-Angled Triangle:\n");
```

```
        decreasingStar(n);
```

```
        break;
```

```
    case 2:
```

```
        printf("\nDecreasing Natural-Number Pattern:\n");
```

```
        decreasingNatural(n);
```

```
        break;
```

```
    case 3:
```

```
        printf("\nDecreasing Repeated-Number Pattern:\n");
```

```
        decreasingRepeated(n);
```

```
break;
```

```
case 4:
```

```
printf("\nReverse-Number Triangle:\n");
```

```
reverseNumber(n);
```

```
break;
```

```
default:
```

```
printf("Invalid choice.\n");
```

```
}
```

```
return 0;
```

```
}
```

Test Case Table:

Test Case	Choice	Input	Expected Output
1	1	4	* * * * * * * * * *
2	2	4	1 1 1 1
3	3	4	4 3 2 1 3 2 1 2 1 1
4	5	4	Invalid choice
5	1	0	Invalid number of rows

Compilation and Execution:

The following test cases have been compiled and executed successfully.

Test Case 1:

```
(kali@kali)-[~]  
$ ./a.out  
----- DECREASING PATTERNS -----  
1. Decreasing Right-Angled Triangle  
2. Decreasing Natural-Number Pattern  
3. Decreasing Repeated-Number Pattern  
4. Reverse-Number Triangle  
Enter your choice: 1  
Enter number of rows: 4  
  
Decreasing Right-Angled Triangle:  
* * * *  
* * *  
* *  
*
```

Test Case 2:

```
(kali@kali)-[~]  
$ ./a.out  
----- DECREASING PATTERNS -----  
1. Decreasing Right-Angled Triangle  
2. Decreasing Natural-Number Pattern  
3. Decreasing Repeated-Number Pattern  
4. Reverse-Number Triangle  
Enter your choice: 2  
Enter number of rows: 4  
  
Decreasing Natural-Number Pattern:  
1  
1  
1  
1
```

Test Case 3:

```
(kali㉿kali)-[~]  
$ ./a.out  
----- DECREASING PATTERNS -----  
1. Decreasing Right-Angled Triangle  
2. Decreasing Natural-Number Pattern  
3. Decreasing Repeated-Number Pattern  
4. Reverse-Number Triangle  
Enter your choice: 3  
Enter number of rows: 4  
  
Reverse-Number Triangle:  
4321  
321  
21  
1
```

Test Case 4:

```
(kali㉿kali)-[~]  
$ ./a.out  
----- DECREASING PATTERNS -----  
1. Decreasing Right-Angled Triangle  
2. Decreasing Natural-Number Pattern  
3. Decreasing Repeated-Number Pattern  
4. Reverse-Number Triangle  
Enter your choice: 5  
Enter number of rows: 4  
Invalid choice.
```

Test Case 5:

```
(kali㉿kali)-[~]  
$ ./a.out  
----- DECREASING PATTERNS -----  
1. Decreasing Right-Angled Triangle  
2. Decreasing Natural-Number Pattern  
3. Decreasing Repeated-Number Pattern  
4. Reverse-Number Triangle  
Enter your choice: 1  
Enter number of rows: 0  
Invalid number of rows.
```